

Aryan Miriyala

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PROFESSIONAL SUMMARY

Software engineer and CS graduate student focused on **data**, **AI**, and intelligent systems, with experience building **Python**, **PySpark**, **Spark/Hadoop**, **Docker**, and **AWS** data workflows. Background includes production cloud ETL, analytics-ready datasets, ML experimentation, RAG/search systems, and reliable internal software.

EDUCATION

Bowling Green State University

Master of Science in Computer Science, GPA: 4.00/4.00

Bowling Green, OH
Aug. 2024 – Aug. 2026

Bowling Green State University

Bachelor of Science in Computer Science, GPA: 4.00/4.00

Bowling Green, OH
Jan. 2021 – Apr. 2024

EXPERIENCE

Artificial Intelligence & Computer Science Intern

SmartSolve Industries

May 2026 – Present
Bowling Green, OH

- Built secure internal software with **Next.js**, **PostgreSQL**, **Drizzle ORM**, **SSO**, and auth middleware, organizing onboarding and operations workflows around sensitive employee data.
- Created a cross-platform **Docker** devcontainer for **WSL2/Colima**, standardizing reproducible local setup for AI-assisted internal software development.

Data Engineering Intern

American Association of Insurance Services

May 2025 – Aug. 2025
Lisle, IL

- Automated a manual **SQL** billing workflow with **Python**, **PySpark**, and **AWS Glue Workflows**, processing **20+ TB** of golden-table insurance data for **700+ member companies**.
- Built cloud ETL and partitioned **S3** pipelines that replaced **160+** legacy Pentaho jobs for **Semarchy MDM**, improving maintainability and standardizing scalable data movement.
- Profiled **160+ MySQL/Oracle/Impala** tables through **JDBC** with **Python**, **Pandas**, and **SQL**, mapping source-system similarities into **25** MDM domains.
- Implemented reverse-ETL, data-validation, and custom **AWS IAM** access patterns to support controlled self-service data access and reliable downstream analytics workflows.

Software Engineering Intern

American Association of Insurance Services

May 2023 – May 2024
Lisle, IL

- Generated **1,000+** production SQL tables across **MySQL**, **Oracle**, and **PostgreSQL** with **Python/JSON** workflows, supporting standardized insurance data models across **10+** lines.
- Built **Python/Pandas** data-processing workflows with multiprocessing, fuzzy matching, and hashing to clean large CSV datasets used in insurance data-standardization pipelines.
- Automated database validation with **AWS Lambda** triggers that tokenized PII using **hashlib/boto3** and ingested structured JSON into **S3** for downstream data workflows.
- Implemented **RBAC** and **JWT** patterns in a **React/Node.js** modernization effort, replacing broad employee-wide access with role-specific controls for about **110** users.

Software Engineering Intern

Alliance for Paired Kidney Donation

May 2022 – Aug. 2022
Toledo, OH

- Improved an **AWS-hosted Lucee/CFML** healthcare workflow platform with anti-CSRF protection, IP-aware audit logging, protected routing, and data-entry reliability fixes.
- Modernized legacy forms with **HTML5**, JavaScript fallbacks, themed CSS, and reusable modal/tab patterns to improve maintainability for internal clinical operations workflows.

PROJECTS

HealthTrend | *Apache NiFi, Kafka, Hadoop HDFS, Spark, PySpark, Docker*: Built a Dockerized healthcare data pipeline routing CSV patient records into HDFS and JSON IoT health events into Kafka, then processing both sources with Spark/PySpark for validation and downstream analytics.

MarioRL Agent Comparison | *Python, PyTorch, Stable-Baselines3, Gym, TensorBoard*: Compared PPO and DQN agents with frame preprocessing, checkpoints, logs, clips, and performance graphs for reward, episode length, and training speed.

FalconGraph Search | *Python, Flask, OpenAI API, RAG, React, C++*: Built a campus search prototype with document ingestion, chunking, vector retrieval, semantic search, source-aware summaries, and graph-based answer tracing.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript, C++/C, C#, Lucee/CFML

AI/ML: PyTorch, TensorFlow, scikit-learn, Stable-Baselines3, TensorBoard, HuggingFace, Sentence Transformers, RAG, embeddings, vector search, semantic retrieval, LangChain, OpenAI API, Mistral AI

Data, Cloud & Distributed Systems: PySpark, Spark, Hadoop HDFS, MapReduce, Kafka, Apache NiFi, AWS Glue, S3, Lambda, IAM, ETL, reverse ETL, data validation, data profiling, partitioned S3 pipelines

Platforms & Tools: Docker, Docker Compose, Git, Linux, PostgreSQL, MySQL, Oracle, Impala, MongoDB, JDBC, React, Node.js